

Solutions – Practice Problem Set 1

ECON 204 – Contemporary Economic Issues

Winter 2026

Part I: Inflation, Cost of Living, and Monetary Policy

Question 1: Measuring Inflation

CPI last year = 152.0, CPI this year = 158.1.

(1) Inflation rate

$$\pi = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} \times 100 = \frac{158.1 - 152.0}{152.0} \times 100$$

Compute the numerator:

$$158.1 - 152.0 = 6.1$$

Now divide:

$$\frac{6.1}{152.0} \approx 0.0401316$$

Convert to percent:

$$0.0401316 \times 100 \approx 4.01\%$$

Answer: Inflation $\approx 4.01\%$.

(2) Real wage change

Nominal wage growth = 3.5%. Approximate real wage growth:

$$\Delta w^{real} \approx \Delta w^{nominal} - \pi = 3.5\% - 4.01\% = -0.51\%$$

Answer: Real wage falls by about 0.5%.

(3) Purchasing power interpretation

If inflation exceeds nominal wage growth, the worker's pay buys fewer goods and services than before. Even with a raise, the worker is effectively worse off in real terms.

Question 2: Cost of Living and Household Budgets

Budget shares: housing 30%, food 20%, transport 15%, utilities 10%, other 25%. Price changes: housing 8%, food 5%, transport 4%, utilities 6%, other 2%.

(1) Household-specific inflation rate

This is a weighted average:

$$\pi^{HH} = \sum w_i \pi_i$$

Convert shares to decimals:

Housing $w_H = 0.30$, $\pi_H = 0.08$

Food $w_F = 0.20$, $\pi_F = 0.05$

Transport $w_T = 0.15$, $\pi_T = 0.04$

Utilities $w_U = 0.10$, $\pi_U = 0.06$

Other $w_O = 0.25$, $\pi_O = 0.02$

Compute each contribution:

Housing: $0.30 \times 0.08 = 0.024$

Food: $0.20 \times 0.05 = 0.010$

Transport: $0.15 \times 0.04 = 0.006$

Utilities: $0.10 \times 0.06 = 0.006$

Other: $0.25 \times 0.02 = 0.005$

Sum:

$$0.024 + 0.010 + 0.006 + 0.006 + 0.005 = 0.051$$

Convert to percent:

$$0.051 \times 100 = 5.1\%$$

Answer: Household inflation 5.1%.

(2) Why differs from CPI

The CPI is based on an “average” basket. A household’s basket differs because:

Different expenditure weights (e.g., renters vs homeowners; families vs singles).

Different local prices (housing and transport are highly regional).

Some households are more exposed to items with high inflation (like rent).

(3) Who most affected when housing rises fastest

Households with high housing shares: renters, low- and middle-income households, single parents, and households in tight metro rental markets. Since housing is a necessity, they have less ability to substitute away.

Question 3: Real Interest Rates and Monetary Policy

Nominal interest rate $i = 5.5\%$. Expected inflation $\pi^e = 2.8\%$.

(1) Real interest rate

Using Fisher approximation:

$$r \approx i - \pi^e = 5.5\% - 2.8\% = 2.7\%$$

Answer: 2.7%.

(2) How real rate affects decisions

A higher real interest rate:

Raises the true cost of borrowing $\rightarrow$ discourages consumption financed by credit (cars, housing) and discourages business investment.

Increases the return to saving $\rightarrow$ encourages saving.

Generally reduces aggregate demand over time.

(3) How a policy rate hike lowers inflation (channels)

Textbook transmission mechanism:

Interest rate channel: borrowing costs rise $\rightarrow$ consumption and investment fall.

Housing channel: mortgages more expensive $\rightarrow$ housing demand slows $\rightarrow$ house price growth moderates; also reduces related spending.

Exchange rate channel (Canada): higher rates can appreciate CAD $\rightarrow$ imports cheaper $\rightarrow$ lowers inflation.

Expectations/credibility: credible tightening anchors inflation expectations $\rightarrow$ wage/price setting becomes less inflationary.

Question 4: Inflation Expectations

Expected inflation rises from 2% to 5%.

(1) Wage bargaining

Workers seek higher nominal wages to maintain purchasing power. They may request wage increases closer to expected inflation plus productivity growth.

(2) Firm response

Firms anticipate higher wage costs and may raise prices preemptively, especially in sectors with pricing power.

(3) Expectations-driven persistence

If wages rise because inflation is expected to rise, and prices rise because wages rose, expectations can become self-fulfilling (a wage–price dynamic). Even if the original shock fades, inflation can persist because expectations influence current contracts.

Part II: Income Inequality

Question 5: Mean and median income

Incomes: 20,000, 30,000, 45,000, 70,000, 135,000.

(1) Mean

Sum:

$$20,000 + 30,000 = 50,000$$

$$50,000 + 45,000 = 95,000$$

$$95,000 + 70,000 = 165,000$$

$$165,000 + 135,000 = 300,000$$

Mean:

$$\bar{y} = \frac{300,000}{5} = 60,000$$

Answer: Mean = \$60,000.

(2) Median

With 5 observations, the median is the 3rd income (ordered list):

$$\text{Median} = 45,000$$

Answer: Median = \$45,000.

(3) Why mean differs

The high income (\$135,000) pulls the mean upward. The median is less sensitive to extremes, so it can better represent the “typical” household in a skewed distribution.

Question 6: Income shares

Total income = 300,000.

(1) Total income

$$Y = 300,000$$

(2) Share of top household

Top household income = 135,000.

$$\text{Top share} = \frac{135,000}{300,000} \times 100 = 45\%$$

Answer: 45%.

(3) Share of bottom two households

Bottom two: $20,000 + 30,000 = 50,000$.

$$\text{Bottom two share} = \frac{50,000}{300,000} \times 100 = 16.67\%$$

Answer: 16.67%.

(4) Interpretation

The top household receives nearly half of all income in this small economy, indicating high concentration. Bottom two receive only about one-sixth, highlighting inequality.

Question 7: Policy and redistribution

Transfer \$8,000 to each of two lowest; tax \$16,000 on highest.

Original incomes:

A: 20,000

B: 30,000

C: 45,000

D: 70,000

E: 135,000

(1) New incomes

A:

$$20,000 + 8,000 = 28,000$$

B:

$$30,000 + 8,000 = 38,000$$

E:

$$135,000 - 16,000 = 119,000$$

C and D unchanged.

New set:

A: 28,000

B: 38,000

C: 45,000

D: 70,000

E: 119,000

(2) New mean

Check total: transfers are funded by tax, so total income should remain 300,000. Sum to verify:

$$28,000 + 38,000 = 66,000$$

$$66,000 + 45,000 = 111,000$$

$$111,000 + 70,000 = 181,000$$

$$181,000 + 119,000 = 300,000$$

Mean:

$$\bar{y} = \frac{300,000}{5} = 60,000$$

Answer: Mean remains \$60,000.

(3) Inequality up or down

Inequality decreases because the bottom rises and the top falls. For example, top-to-bottom ratio:

Before:

$$135,000/20,000 = 6.75$$

After:

$$119,000/28,000 \approx 4.25$$

The distribution is less spread out.

Question 8: Poverty threshold \$28,000

(1) Below poverty line

Using original incomes:

20,000 < 28,000 → poor

30,000, 45,000, 70,000, 135,000 → not poor

Answer: Only Household A is below.

(2) Poverty rate

$$\frac{1}{5} \times 100 = 20\%$$

Answer: 20%.

(3) One limitation

A poverty line is a sharp cutoff: someone earning \$27,900 is “poor” while \$28,100 is “not poor,” though living standards are nearly identical. It also may not adjust well for local cost-of-living differences.

Part III: Housing Affordability and Real Estate Markets

Question 9: Price-to-income ratio

Income = 95,000, home price = 760,000.

(1) PTI

$$PTI = \frac{760,000}{95,000} = 8$$

Answer: 8.

(2) Interpretation

A PTI of 8 means the home costs eight years of gross income – typically considered severely unaffordable, especially once interest costs and down payments are considered.

(3) Why incomplete

PTI ignores mortgage rates, down payment constraints, property taxes, condo fees, and household composition; also ignores whether the house is larger/better quality than typical.

Question 10: Mortgage payments and interest rates

Loan $P = 500,000$, amortization 25 years = 300 months.

Monthly payment formula:

$$M = P \cdot \frac{r(1+r)^n}{(1+r)^n - 1}$$

where r is monthly interest rate and $n = 300$.

We compute two cases.

Case 1: 2.5% annual

Monthly rate:

$$r_1 = \frac{0.025}{12} = 0.00208333$$

Compute $(1 + r_1)^{300} \approx 1.867$ (approx). Then:

$$M_1 \approx 500,000 \cdot \frac{0.00208333 \cdot 1.867}{1.867 - 1}$$

Numerator: $0.00208333 \times 1.867 \approx 0.003889$

Denominator: 0.867

Fraction: $0.003889/0.867 \approx 0.004484$

$$M_1 \approx 500,000 \times 0.004484 \approx 2,242$$

So payment is about \$2,240/month.

Case 2: 5.5% annual

Monthly rate:

$$r_2 = \frac{0.055}{12} = 0.00458333$$

Compute $(1 + r_2)^{300} \approx 3.94$ (approx).

$$M_2 \approx 500,000 \cdot \frac{0.00458333 \cdot 3.94}{3.94 - 1}$$

Numerator: $0.00458333 \times 3.94 \approx 0.01806$

Denominator: 2.94

Fraction: $0.01806/2.94 \approx 0.00614$

$$M_2 \approx 500,000 \times 0.00614 \approx 3,070$$

So payment is about \$3,070/month.

(1) Change

$$\Delta M \approx 3,070 - 2,240 \approx 830$$

Answer: Monthly payment rises by about \$800–\$850/month.

(2) Why affordability worsens even if prices stabilize

Higher interest rates raise the monthly payment for any given loan size, and at renewal they raise cash-flow burdens. Even if the house price stops rising, households can still be “priced out” by monthly payments.

Question 11: Rental affordability

Income = 3,800/month. Rent rises from 1,200 to 1,850.

(1) Rent-to-income ratios

Before:

$$\frac{1,200}{3,800} = 0.3158 \Rightarrow 31.58\%$$

After:

$$\frac{1,850}{3,800} = 0.4868 \Rightarrow 48.68\%$$

Answer: About 31.6% → 48.7%.

(2) Threshold

A common benchmark is 30% (and sometimes 50% for “severe burden”).

(3) Link to inequality

Rising rents hit lower- and middle-income renters hardest because housing is a necessity. It reduces ability to save, invest in education, or accumulate wealth-widening the gap between renters and homeowners.

Question 12: Supply elasticity and prices (explanations)

(1) Restrictive zoning

Restrictive zoning makes supply less responsive (more inelastic). When demand rises, fewer new units are built, so prices and rents increase more.

(2) Increased immigration

Raises population and household formation, increasing demand for rentals and ownership. With slow supply response, affordability worsens through higher rents/prices.

(3) New transit infrastructure

Transit increases accessibility and raises desirability of nearby locations. It can increase housing demand in those areas (prices rise). Over time, if zoning allows densification around transit, supply can increase and moderate long-run price growth.

(4) Decline in mortgage interest rates

Increases borrowing capacity; demand shifts outward; prices rise, especially when supply is constrained.

Part IV: Integrated Policy Analysis

Question 13: Inflation and housing (combined mechanisms)

High inflation often leads central banks to raise interest rates. Higher rates:

Slow house price growth by reducing mortgage borrowing capacity and cooling demand.

Increase mortgage payments for variable-rate borrowers immediately and for fixed-rate borrowers at renewal.

Raise rental demand because higher payments and stress tests price out marginal buyers, pushing them into renting, which can raise rents.

So, it is possible for house price inflation to slow while renter cost pressures intensify.

Question 14: Inequality and housing wealth

Rising house prices increase wealth inequality because housing is a key asset.

Homeowners vs renters: Owners gain capital gains and build equity; renters do not. Higher prices also raise barriers to entry for renters to become owners.

Older vs younger households: Older cohorts are more likely to already own homes; they benefit from appreciation. Younger households face higher entry prices, larger down payments, and higher debt burdens.

This creates an intergenerational “asset divide.”